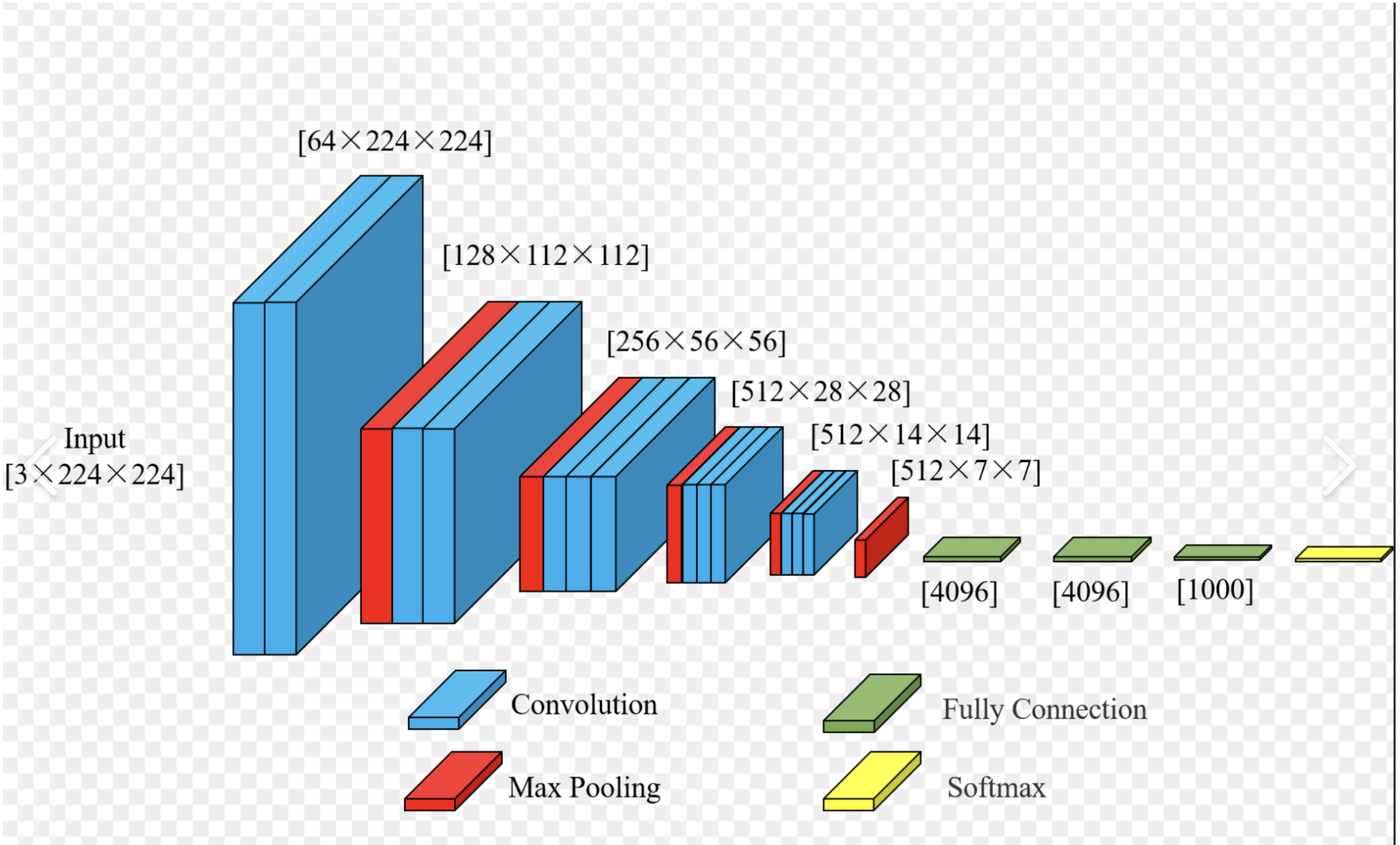

The core contribution of VGG is that it proved that increasing the depth of the network can significantly improve performance.

VGG16 proved that increasing neural network depth consistently improves image classification accuracy, demonstrating through rigorous experiments that going from 8 layers (AlexNet) to 16 layers reduced ImageNet error from 37.5% to 26.5%, and further extending to 19 layers continued to yield marginal gains, thereby establishing depth as a critical factor for model performance. It achieved this by using only 3x3 convolution kernels stacked repeatedly, which kept parameters manageable while allowing deeper architectures, and it showed that with proper techniques like dropout and careful initialization, the optimization problems that previously limited depth could be overcome. However, VGG also revealed the practical limits of simply adding more layers, as its 138 million parameters made it computationally expensive and memory-intensive, ultimately inspiring later architectures like ResNet that used skip connections to train even deeper networks efficiently.